Ofqual QAN	**600/0406/X**
Quartz ID	**1156, 1157**
Qualification	**Level 4 Diploma in Computing. Level 4 Diploma in Computing with Business Management**

Assessment	**Algorithms and Mathematical Concepts for Computing**
Assessment Type	**Examination**
Date	**27th February 2025**
Time	**2.00pm – 5.00pm**

Examination Paper

Answer ALL questions.

Clearly cross out surplus answers.

Time: 3 hours

The maximum mark for this paper is 100.

Any reference material brought into the examination room must be handed to the invigilator before the start of the examination.

Answer ALL questions

Marks

Question 1

This question relates to the Software Development Process and Algorithm:

a) At which stage of the software development lifecycle is the actual coding and integration of the system components carried out to turn the design into a working application? **1**

b) In which stage of the software development lifecycle is the application validated for functionality, usability and security to ensure it meets the defined specifications and user requirements? **1**

c) Explain the difference between an algorithm and a program. **8**

Discuss the purpose of algorithms and programs.

Describe the hierarchy of algorithmic complexity, providing an example for each complexity level.

Total 10 Marks

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Question 2

This question relates to numbers and number systems:

a) Which of the following is **NOT** a valid number system? **1**

i) Binary number system
ii) Decimal number system
iii) Fractional number system
iv) Octal number system

b) Conversion from decimal to _____ involves converting a number from base 10 to base 8. **1**

c) Describe the concept of least common multiple (LCM) **and** explain its significance in number theory. **2**

d) Perform the following conversions **and** show your working.

i) Convert the octal number 54 to binary. Show your working. **2**

ii) Convert the hexadecimal number 1D to decimal. Show your working. **2**

iii) Convert the decimal number 75 to binary. Show your working. **2**

Total 10 Marks

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Question 3

Answer the following questions related to logic and Boolean algebra:

a) Which of the following operators represents the logical ***AND*** operation? **1**

i) $\wedge$
ii) $\vee$
iii) $\neg$
iv) $\rightarrow$

b) Complete the truth table for the Boolean expression: **4**

$$(P \vee Q) \wedge \neg(P \wedge Q)$$

P	**Q**	**P ∧ Q**	**P ∨ Q**	**¬(P ∧ Q)**	**(P ∨ Q) ∧ ¬(P ∧ Q)**
T	T				
T	F				
F	T				
F	F				

[Insert the marking scheme for part (b) here.] ***[Insert***

c) Explain the concept of logical implication in Boolean algebra **and** provide **TWO (2)** examples. Include the truth table for logical implication. **5**

[Insert the marking scheme for part (b) here.] ***[Insert***

Total 10 Marks

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Question 4

Answer the following questions related to Predicate Logic, Set Theory and Venn Diagrams:

a) Which of the following symbols represents the intersection operation in set theory? **1**

i) ⊆
ii) ⊂
iii) ∪
iv) ∩

[Insert the marking scheme for part (b) here.]

b) Explain the concept of set union **and** provide **ONE (1)** example. **2**

[Insert the marking scheme for part (b) here.]

c) Calculate the complement of set A = {1,3,5} relative to the universal set U = {1,2,3,4,5,6} **and** explain the process. **2**

d) Describe the power set of a set? **3**

Calculate the power set of A = {x,y} **and** provide the correct representation.

[Insert the marking scheme for part (b) here.]

e) Explain the key difference between propositional logic and predicate logic in terms of expressiveness. **2**

[Insert the marking scheme for part (b) here.]

[Insert the marking scheme for part (b) here.]

Total 10 Marks

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Question 5

Answer the following questions related to statistics:

a) Explain the concept of variance. **2**

Provide **ONE (1)** example of a scenario where it is appropriate to use variance.

b) Write the formula for calculating variance. **3**

Calculate the variance for the following data set: 4, 8, 6, 5, 9. Show your working.

c) Describe the concept of kurtosis in statistics **and** explain its importance in analysing data distribution. **2**

d) Describe median **and** explain how it is calculated. **3**

Determine the median for the following data set: 12, 7, 15, 10, 5. Show your working.

Total 10 Marks

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Question 6

Answer the following questions related to Data Structures and Arrays:

a) Which of the following is **NOT** a characteristic of an array? **1**

i) Stores elements of the same data type.
ii) Allows random access to elements.
iii) Can automatically sort its elements.
iv) Elements are stored in contiguous memory locations.

b) Define the data type "queue". Explain **TWO (2)** operations on a queue and provide ONE (1) example of an application where it would be used. **4**

c) Write pseudocode to dequeue an element from a queue. Assume the existence of a function called isEmpty(queue) that checks if the queue is empty. **5**

Total 10 Marks

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Question 7

Answer the following questions related to Data Structures:

a) Which of the following data structures stores elements in nodes where each node contains a reference (or pointer) to the next node in the sequence? **1**

i) Array
ii) Queue
iii) Stack
iv) Linked List

b) In a stack, elements are added to the _____ and removed from the _____. **2**

c) Consider a queue initially containing the elements 10, 20, 30, 40. Assume 10 is at the front of the queue. Perform the following operations one after the other: **3**

1. Add the number 50.
2. Dequeue two elements.
3. Add the numbers 60 and 70.

Draw out the queue after each operation. Show your working.

d) Provide **TWO (2)** computing-related examples where a stack data structure would be appropriately used. Explain your answer for each example. **4**

Total 10 Marks

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Question 8

Answer the following questions related to algorithm efficiency and searching techniques:

a) Which of the following best describes the concept of asymptotic analysis in algorithms? **1**

i) It measures the exact runtime of an algorithm for specific input sizes.
ii) It evaluates the algorithm's performance for very large input sizes.
iii) It calculates the number of operations for the smallest input size.
iv) It determines the memory usage of an algorithm in all cases.

b) Using the Big-O notation state the worst-case time complexity for a linear search algorithm. **1**

c) Write pseudocode for a function that implements an iterative linear search algorithm to find a target value in an array. The function should take the array and target value as inputs and return the index of the target value, or -1 if the target is not found. **6**

d) Explain the efficiency of binary search with respect to its time complexity. Provide **ONE (1)** scenario where binary search is more suitable than other search algorithms. **2**

Total 10 Marks

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Question 9

Answer the following questions related to sorting algorithms and recursion:

a) Which sorting algorithm repeatedly divides the input array into smaller subarrays, sorts each subarray and then combines them to produce the final sorted array? **1**

i) Bubble Sort
ii) Quick Sort
iii) Merge Sort
iv) Selection Sort

b) State what is meant by a recursive base case in programming **and** explain its importance in ensuring the termination of a recursive function. Provide **ONE (1)** example scenario where the base case is crucial. **3**

c) Fill in the blanks in the pseudocode for computing the sum of numbers from 1 to n using recursion: **4**

```
Function sumToN(n):
  if _____:
     return _____
  return _____ + _____ (n - 1)
End function
```

d) Explain how recursion can be used to solve the Tower of Hanoi problem. **2**

Total 10 Marks

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Question 10

Answer the following questions related to tree data structures:

a) Which of the following is ***TRUE*** about a binary search tree (BST)? **1**

i) The left subtree of a node contains only nodes with values greater than the node.

ii) The right subtree of a node contains only nodes with values smaller than the node.

iii) For each node, its left and right subtrees are binary search trees.

iv) The root of the tree contains the smallest value in the entire tree.

b) In a balanced binary tree, the height difference between the left and right subtrees of any node is at most _____. **1**

c) Construct the binary search tree (BST) for the following sequence of numbers: 50, 30, 20, 70, 40, 80, 60. Ensure that 50 is the root. **4**

d) Using the binary search tree (BST) constructed in Question c), explain the steps to search for the value 25. Describe each comparison and movement until the value is found or determined to be absent. **4**

Total 10 Marks

End of paper

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